

Caden Whalen

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Objective

Motivated Aerospace and Electrical Engineering major with a strong skill set in communication and problem-solving, specializing in multidisciplinary work and avionics. Hardworking and adept at managing complex projects and collaborating with others across technical teams. Founder of a highly specialized microsatellite technology organization centered around providing and testing cost-effective, open-source design solutions. Seeking internship opportunities in the general avionics field starting May 2027.

Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Aerospace Engineering, GPA 4.00

Bachelor of Science in Electrical Engineering, GPA 4.00

August 2025 – Present

Expected Graduation, May 2029

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Skills

Programming: Assembly, C, Glowscript, Java, MATLAB, Python

Platforms: Altium Designer, Linux (Kali/Ubuntu), OpenRocket, VirtualBox

Hardware: Ammeters, PCB Prototyping, Raspberry Pi, Strain Gauges, Voltmeters

Software: GitHub, Siemens NX, SolidWorks

Professional Organizations: American Institute of Aeronautics and Astronautics

Communication: Conflict Resolution, Mentorship, Public Speaking, Technical Proposal Writing

Languages: English (fluent), Spanish (conversational), Japanese (beginner)

Experience

Georgia Tech Orbital Initiative | Atlanta, GA

March 2026 – Present

Founder, President

Student organization aiming to advance access to space technology through the design, development, and testing of cost-effective, open-source microsatellites.

- Identified a lack of support and opportunity for students interested in microsatellite technology; founded and developed a specialized organization currently in the process of creating 3 subsystem designs; estimated to reduce traditional costs by 40%.
- Positioned to become officially recognized by Georgia Tech in Fall 2026.

Yellow Jacket Space Program | Atlanta, GA

January 2026 – Present

Future Responsible Engineer

Student organization working to design, construct, and fly a liquid-fueled rocket to the Karan Line (100 km), the official edge of space.

- Responsible for addressing a faulty sensor; designing, creating, and integrating a sensor controller PCB with proper electrostatic discharge protection, reverse polarity protection, and proper filtering; resulting in the restoration of full sensor functionality.
- Positioned to lead an avionics project by August 2026.

Projects

Miniature Ant Robot for Search and Rescue | Engineering Graphics

August - December 2025

Frame & Battery Lead

Team-based project tasked with designing a miniature robot capable of locating and aiding in the rescue of victims trapped in collapsed buildings during natural disasters.

- Responsible for designing the carbon fiber frame and retractable battery mechanism as well as integrating a servo-powered 360-degree camera, IoT audio network, and LED lights, ultimately resulting in over 40 unique, interlocking components.

Relevant Coursework

Circuit Analysis: Mastered the characterization of linear circuits using Thevenin-Norton theorems, Node Voltage Analysis, and superposition; phasor analysis, Bode plots, and operational amplifiers; utilized Laplace transforms to analyze two-port systems.

Digital System Design: Mastered the underlying computer system and digital design principles; switch and gate design, Boolean algebra, number systems, arithmetic, and storage elements; instruction set architecture, assembly language; memory organization.

Thermodynamics & Fluids Fundamentals: Mastered the analysis and applications of fundamental conservation equations and how they govern fluid and thermodynamic systems; converging-diverging nozzles and compressible flow; aerospace design applications.